Sahaj Bhadja

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Summary

Graduate software engineer with 2+ years building **C++/Python optimization tooling**, **high-performance data workflows**, and **algorithm-driven systems**, with strong foundations in concurrency, memory efficiency, and applied ML experimentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **performance-focused backend engines** for 5K+ monthly jobs and 500+ users, reducing query latency from **1.2s to 350ms** through algorithmic optimization, indexing strategy, and efficient data path redesign.
- Implemented asynchronous orchestration workflows with explicit job-state transitions, failure isolation, and retry controls, improving reliability and enabling scalable execution patterns for high-volume computational pipelines.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Developed **Python benchmarking pipelines** over 100K+ trajectories, designing reproducible experiments and analytical metrics to evaluate behavior under varied constraints and optimize downstream algorithmic decision quality.
- Partnered with multidisciplinary engineers and researchers to translate experimental findings into implementation guidance, improving iteration speed on complex optimization tasks and reducing ambiguity in technical design tradeoffs.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **algorithmic ETL workflows** using Python/Java and AWS Glue/S3, improving response times by **30%** while scaling distributed processing and preserving correctness across high-volume integration pipelines.
- Maintained **20+ automated unit/integration suites** with Jenkins, PyTest, and Mockito, improving CI efficiency by **68%** and reducing technical debt by **20%** through disciplined quality gates.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Built high-throughput full-stack services handling 10K+ daily requests, optimizing concurrent processing paths and model-inference workflows for stable performance under evolving product requirements and short iteration cycles.
- Accelerated release quality through CI/CD automation and regression controls, reducing turnaround time by **60%** while maintaining robust behavior across integrated APIs, data pipelines, and user-facing components.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Developed REST/WebSocket modules supporting eight workflows and 2M+ synchronized records, improving real-time consistency and reducing state drift in distributed application interactions under sustained load.
- Automated Docker/Jenkins build and deployment workflows, reducing release time by **50%** and improving repeatability of performance-sensitive test and rollout operations across Linux-based engineering environments.

Skills

Languages: **C++14+**, **Python**, **Java**, SQL, Tcl/Perl familiarity
Algorithms & Performance: optimization engines, concurrency, efficient memory/I/O usage, multithreaded design
Engineering: OOP, data structures, CI/CD, Jenkins, Git, automated testing, production debugging
ML/Research: large-scale experimentation, model benchmarking, AI-assisted workflows, reproducible evaluation pipelines

Projects

CPU Scheduling Simulator | *React, JavaScript, OS Algorithms*

- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling algorithms with interactive Gantt visualization, reducing average turnaround by **25%** and improving debugging efficiency by **20%** across 1K+ scenarios.

CaptionCraft - Image Caption Generator | *Python, Transformers, Vision Models*

- Benchmarked BLIP, GIT, and ViT-GPT2 architectures on 10K images and 100K captions, increasing caption accuracy from **81% to 96%** through systematic tuning and algorithm-level validation experiments.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Designed a schema-aware SQL generation workspace for 10+ databases with structured validation and execution checks, improving algorithmic correctness and reducing manual troubleshooting in complex data processing workflows.